Sadguru Balumama Shikshan Prasarak Mandal's
K.P.PATIL INSTITUTE OF TECHNOLOGY
(DTE CODE : 6814) (MSBTE Code : 1661)
Approved by AICTE, DTE Mumbai &Govt. of Maharashtra,Affiliated to MSBTE Mumbai

Internships Daily Diary(2025-26)
Department Of Artificial Intelligence & Machine Learning Engineering

Program Code :AI5K
Name of the Student : Sanika Dhanaji Chougale
Name of the Mentor (Faculty) : Mrs. P.M. Kumbhar
Name of the Mentor (Industry): Mr. Abhijit Gatade
Enrollment Number : 23213500044

Week	Day & Date	Discussion Topics/Activity	Details of Work Allotted Till Next Session /Corrections Suggested/Faculty Remarks	Signature of Industry Mentor
Week 7	Mon, Date 14/07/25	♦ Dataset Loading & Data Cleaning	imported data and removed missing /duplicate rows	
	Tue, Date 15/07/25	♦ EDA and Visualization Insights	plotted graphs to explore data relationships	
	Wed, Date 16/07/25	♦ Model Selection and feature Encoding	Trained models and handled Categorical features	
	Thu, Date 17/07/25	♦ Model Tuning & pipeline Creation	improved accuracy with GridSearch CV pipeline	
	Fri, Date 18/07/25	♦ Streamlit App Development Started	Designed UI & Connected prediction model	
	Sat, Date 19/07/25	♦ Final Touches & Documentation	Validated inputs and wrote README file	

Internships Daily Diary (2025-26)

Week No: 07 **Day:** Monday **Date:** 14/07/25

"Dataset Loading & Data Cleaning"

Today, I began by loading the car price dataset into the working environment using pandas. I inspected the dataset structure to understand the available features and target variable.

I checked for missing values, duplicate entries, and incorrect data types. After analysis, I removed rows with missing values and dropped duplicate records to ensure data quality. I also standardized column names for consistency and converted apporopriate features to correct data types.

This cleaning step helped ensure a solid foundation for further processing and model building.

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Internships Daily Diary (2025-26)

Week No: 07 **Day:** Tuesday **Date:** 15/07/25

"EDA & Visualization Insights"

I performed Exploratory Data Analysis (EDA) to understand the relationships between variables. Using Matplotlib and seaborn, I plotted graphs such as histograms, box plots, heatmaps, and scatter plots to visualize trends and Correlations.

I identified which numerical features affected the car price the most and how Categorical features like brand or fuel type influenced the outcome. These insights guided features selection for model building. I also noted any outliers and began thinking about feature transformation if required.

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Internships Daily Diary (2025-26)

Week No: 07 **Day:** Wednesday **Date:** 16/07/25

"Model Selection & Feature Encoding"

On day, I began working on Converting categorical data into numerical form using techniques like one-Hot-Encoding and Label Encoding. I split the data into training and test sets. I experimented with multiple regression models such as Linear Regression, Random forest Regressor, and XGBoost.

I compared their initial performances using default parameters! I also evaluated models using metrics like R^2 Score, Mean Absolute Error (MAE), and Root Mean Square Error. This step helped shortlist the most promising model for further tuning.

Internships Daily Diary (2025-26)

Week No: 07 **Day:** Thursday **Date:** 17/07/25

"Model Tuning & Pipeline Creation"

Todays was focused on improving model performance. I used Grid Search -CV to perform hyperparameter tuning on the selected models. By definings a parameter grid and applying cross-validation, I found the optimal parameters that improved prediction accuracy. I then created a scikit-learn pipeline to automate the entire process - from preprocessing (scaling, encoding to prediction. This made the workflow modular, clean, and reproducible. final models were saved using joblib for future use.

Internships Daily Diary (2025-26)

Week No: 07 **Day:** friday **Date:** 18/07/25

"Streamlit App Development Started"

I initiated the development of the web-based user interface using streamlit. The app was designed to take user input like car features mileage, brand, fuel type, and output the predicted car price. I linked the trained model into the backend of the app and added input widgets to collect data from users. A clean and simple layout was chosen to keep the interface user-friendly.

I Successfully connected the model and displayed predictions based on real-time input.

Internships Daily Diary (2025-26)

Week No: 07 **Day:** Saturday **Date:** 19/07/25

"Final Touches & Documentation"

On the final day of the week, I completed the last round of testing and improvements. I Validated the app inputs to ensure proper data types and value ranges The README file was written to provide a clear overview of the project, including the problem statement, steps followed, model accuracy, libraries used, and how to run the app. I also organized the project folder structure and pushed the final version to Github.

All tasks for the car price prediction project were successfully Completed.